



Roadblocks to implementing Monte–Carlo method

- In the last lesson, you learned how to approximate an integral using the Monte–Carlo method.
- We assumed that the integrand could be factored into the specific form:

$$\mu = \int_{-\infty}^{\infty} g(x)f(x) \, dx,$$

where $f(x)$ has all of the properties of a pdf.

- Two possible roadblocks to using the Monte–Carlo method:
 1. What if we cannot factor the integrand into the product of a function and a pdf?
 2. What if we cannot generate random values from the pdf $f(x)$?
- We know how to generate pseudo-random samples from a uniform distribution (`rand`) or a Gaussian distribution (`randn`) and perhaps a few others.
 - But, generating pseudo-random samples from a general distribution is difficult even in 1-d, and not really feasible for a high-dimensional problem.



Do these roadblocks impede our progress?

- “Roadblock 1” is not a problem. In the prediction step, the integral we must evaluate is already in the right form. That is,

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) \, dx_{k-1}$$

does factor into a product of a function $g(x_{k-1})$ and a pdf $f(x_{k-1})$:

$$g(x_{k-1}) = f(x_k | x_{k-1}) \quad \text{and} \quad f(x_{k-1}) = f(x_{k-1} | \mathbb{Z}_{k-1}).$$

- But “roadblock 2” is a problem. To compute this integral using the Monte–Carlo method, we must draw independent random numbers from the pdf $f(x_{k-1} | \mathbb{Z}_{k-1})$.
 - Unless this pdf is uniform or Gaussian, we don’t know how to do this.
 - We may be able to *evaluate* the pdf at specific points x_{k-1} (probably not, but we’ll worry about that later) but we can’t draw independent random samples from it.
- We *can* generate uniform or Gaussian pseudo-random numbers—how to use this?



Introducing the “importance density”

- We define an importance density $q(x)$, which is a known pdf from which we are able to draw random samples.
- Then, if we cannot draw random values from $f(x)$ to evaluate:

$$\mu = \int g(x)f(x) \, dx,$$

we could instead consider rewriting the integral as:

$$\mu = \int \frac{g(x)f(x)}{q(x)} \cdot q(x) \, dx \approx \frac{1}{N} \sum_{i=1}^N \frac{g(x^{(i)})f(x^{(i)})}{q(x^{(i)})},$$

where $x^{(i)}$ is the i th sample drawn from the pdf $q(x)$.

- This requires that $q(x)$ have support (*i.e.*, $q(x) > 0$) everywhere $f(x)$ has support.
- Ideally, $q(x)$ is similar to $f(x)$ —weighting which x are most and least *important* when approximating an integral using random samples—but this is not required.



Prior example reconsidered

- Consider the same example as last lesson: we desire to find $\mathbb{E}[y]$, where $y = \cos(x)$ and $x \sim \mathcal{N}(0, \sigma_x^2)$.
- But, suppose that we cannot generate random Gaussian numbers; we have only a uniform pseudo-random number generator $x \sim \mathcal{U}(-R, R)$ for which $f(x) = \frac{1}{2R}$.
- We re-write the problem as: $\mathbb{E}[y] = \int_{-\infty}^{\infty} \frac{\cos(x) \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(\frac{-x^2}{2\sigma_x^2}\right)}{\frac{1}{2R}} \left(\frac{1}{2R}\right) dx$.
- Comparing to our new integral form, $\mu = \int [g(x)f(x)/q(x)] q(x) dx$, we assign:
 $g(x) = \cos(x)$, $f(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(\frac{-x^2}{2\sigma_x^2}\right)$, and $q(x) = \frac{1}{2R}$.
- So, we will draw N independent uniform RVs $x^{(i)}$ between $-R$ and R and compute:

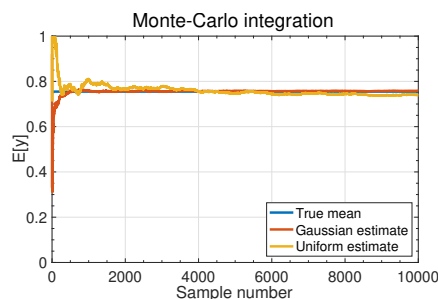
$$\mathbb{E}[y] \approx \frac{2R}{N} \sum_{i=1}^N \cos(x^{(i)}) \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(\frac{-(x^{(i)})^2}{2\sigma_x^2}\right).$$



Octave code for example

```
N = 10000; sx = 0.75; % problem constants
q = 8*sx*(rand([1 N])-0.5); % generate unif RVs
y = cos(q).*exp(-q.^2/(2*sx^2)) ...
    /sqrt(2*pi*sx^2)/(1/(8*sx)); % function evaluations
mu = cumsum(y)./(1:N); % running approximation
plot(1:N, mu); % plot estimate versus N
```

- I chose $q(x) \sim \mathcal{U}(-4\sigma_x, 4\sigma_x)$. Does not span entire support of $f(x)$, but “close enough”.
 - Convergence is slower than before.
 - However, we get the same value in the end.
 - Results will (again) be somewhat different from one trial run to another due to the randomness of the Monte–Carlo method.



Summary

- The integrand of the integral that we desire to evaluate in the prediction step of the particle filter does not factor into a product of $g(x)$ and $f(x)$ where we know how to draw random samples from $f(x)$.
- So, we introduce a new concept: the “importance density” $q(x)$, from whose pdf we can draw random samples. For example, $q(x)$ could be uniform or Gaussian.
- We re-write the integral $\mu = \int_{R_x} g(x)f(x) dx$ as:

$$\mu = \int_{R_x} \frac{g(x)f(x)}{q(x)} \cdot q(x) dx \approx \frac{1}{N} \sum_{i=1}^N \frac{g(x^{(i)})f(x^{(i)})}{q(x^{(i)})},$$

- As long as $q(x)$ has support everywhere that $f(x)$ has support, this will converge to the same solution as $N \rightarrow \infty$.
- An example implemented in Octave code produces similar results to when we could draw random samples from $f(x)$, although convergence was not as fast.